



DELHI PUBLIC SCHOOL NEWTOWN

SESSION: 2023-24

HALF YEARLY EXAMINATION

CLASS: IX
SUBJECT: CHEMISTRY [SET A]

FULL MARKS: 80
TIME: 2 HOURS

General Instructions:

- The paper consists of seven printed pages.
- Section A is compulsory. Attempt any four questions from Section B.
- Answers should be to the point.
- Question numbers should be copied carefully while answering the questions.

SECTION A

(Attempt all questions from this section)

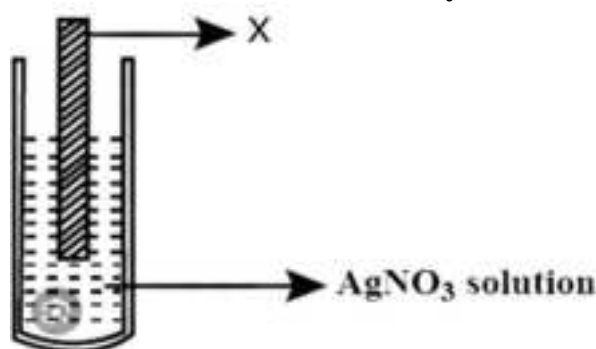
Question 1

Choose one correct answer to the questions from the given options: [15]

- i) The molecular formula of Nessler's reagent is:
- (a) K_2HgI_4
 - (b) K_3HgI_4
 - (c) K_4HgI_4
 - (d) K_2HgI_2
- ii) A metal which reacts with very dilute cold nitric acid to produce hydrogen:
- (a) Sodium
 - (b) Magnesium
 - (c) Calcium
 - (d) Zinc
- iii) A cation which does not have any metallic element in it is:
- (a) Ferrocyanide
 - (b) Phosphate
 - (c) Ammonium
 - (d) Dichromate
- iv) The chemical name of CH_3COONa is:
- (a) Sodium carbonate
 - (b) Sodium formate
 - (c) Sodium oxalate
 - (d) Sodium acetate
- v) The substance used to remove moisture from hydrogen is:
- (a) Anhydrous calcium chloride
 - (b) Concentrated sulphuric acid

- (c) Silver nitrate solution
- (d) Lead nitrate solution

- vi) The salt which on reaction with sodium hydroxide gives a dirty green precipitate is:
- (a) Ferric sulphate
 - (b) Ferrous sulphate
 - (c) Zinc sulphate
 - (d) Copper sulphate
- vii) While performing an experiment Radha dipped a metallic rod X in silver nitrate solution, the colour of the solution turned blue. Identify the metal X:



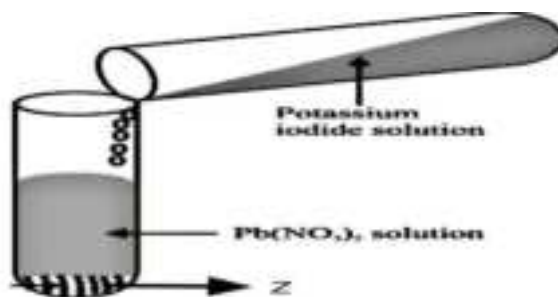
- (a) Silver
 - (b) Iron
 - (c) Copper
 - (d) Zinc
- viii) The ion which contains only 2 electrons:
- (a) H^{2+}
 - (b) H^+
 - (c) H^{2-}
 - (d) H^-
- ix) An atom has 10 neutrons and 10 protons. the valency and type of the element:
- (a) 0, noble gas
 - (b) 4, non-metal
 - (c) 1+, metal gas
 - (d) 2+, metal
- x) The percentage of oxygen in ammonium nitrate [Given that O = 16, N = 14, H = 1] is:
- (a) 16%
 - (b) 60%
 - (c) 32%
 - (d) 48%
- xi) At constant temperature, if the pressure is reduced to 1/6th for a fixed mass of a gas then its volume will become:
- (a) 3 times
 - (b) 1/3 times
 - (c) 6 times

(d) no change

xii) A pungent smelling gas which produces a curdy white precipitate when bubbled through silver nitrate solution is:

- (a) Carbon monoxide
- (b) Carbon dioxide
- (c) Hydrogen
- (d) Hydrogen chloride

xiii) Study the experimental set up given and state the colour and the precipitate Z formed:



- (a) yellow
- (b) bluish white
- (c) dirty green
- (d) dense white

xiv) Which of the following pairs represent two atoms with the same number of neutrons?

- (a) ${}^9\text{F}^{19}$ and ${}_{20}\text{Ca}^{40}$
- (b) ${}^6\text{C}^{13}$ and ${}^7\text{N}^{14}$
- (c) ${}_{11}\text{Na}^{23}$ and ${}_{19}\text{K}^{39}$
- (d) ${}_{27}\text{Co}^{59}$ and ${}_{28}\text{Ni}^{59}$

xv) Absolute zero is:

- (a) 273°C
- (b) -273K
- (c) 0K
- (d) 290°C

Question 2

i) Write the formula for the following compounds:

[5]

- (a) Aluminum nitride
- (b) Calcium bicarbonate
- (c) Sodium phosphate
- (d) Potassium dichromate
- (e) Zinc acetate

- ii) **Balance the following equations:** [5]
- $\text{Pb}_3\text{O}_4 + \text{HCl} \rightarrow \text{PbCl}_2 + \text{H}_2\text{O} + \text{Cl}_2$
 - $\text{Ca}_3\text{N}_2 + \text{H}_2\text{O} \rightarrow \text{Ca(OH)}_2 + \text{NH}_3$
 - $\text{NH}_3 + \text{Cl}_2 \rightarrow \text{NH}_4\text{Cl} + \text{N}_2$
 - $\text{C}_4\text{H}_{10} + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$
 - $\text{Al}_4\text{C}_3 + \text{H}_2\text{O} \rightarrow \text{Al(OH)}_3 + \text{CH}_4$
- iii) **State one appropriate observation for the following chemical reaction taking place:**[5]
- Chlorine is bubbled through an aqueous solution of potassium bromide.
 - Calcium is added to cold water.
 - Moist potassium iodide paper is introduced in a jar of nitrogen dioxide.
 - Zinc nitrate is heated strongly in a test tube.
 - Iron nails are added to an aqueous solution of copper sulphate.
- iv) **Give reasons for the following :** [5]
- Gas laws are not valid at absolute zero.
 - Hydrogen gas is not collected by downward displacement of air.
 - Lime water is not used to distinguish between carbon dioxide and sulphur dioxide.
 - Sodium is not used for the preparation of hydrogen from acids.
 - Isotopes have same chemical properties.
- v) **Zidane took a small amount of a blue crystalline solid [A] in a hard glass test tube which on heating decomposes producing a reddish brown gas [B] with an irritating odour, along with a colourless, odourless neutral gas [C] which rekindles a glowing splinter and leaves behind a black residue [D] .**
- Identify [A], [B], [C] and [D].
 - Give a balanced chemical equation in support of your answer. [5]

SECTION B

(Attempt any four questions)

Question 3

- i) **The composition of two atoms P and Q is given below.**

Atom	Neutrons	Protons	Electrons
P	13	11	11
Q	12	11	11

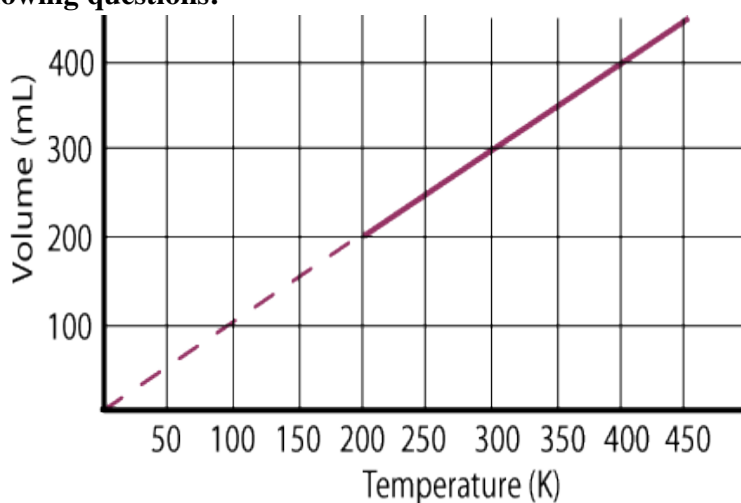
- What is the mass number of P?
- How are P and Q related to each other?
- Are these elements metals or non- metals? [3]

- ii) Divers get 'the bends' if they come up too fast because gas in their blood expands, forming bubbles in their blood. If a diver has 0.05l of gas in his blood under a pressure of 250 atm. What will be the volume of gas in his blood when the pressure decreases to 50 atm? [2]
- iii) Amongst the metals iron, magnesium, zinc and copper which :
 (a) Reacts reversibly with steam.
 (b) Reacts with both acids and alkalis to liberate hydrogen.
 (c) Reacts with steam forming hydrogen and metallic oxide. [3]
- iv) Give a balanced chemical equation for the preparation of the following:
 (a) An acid from sulphur dioxide gas.
 (b) A salt of a trivalent metal and a greenish yellow gas. [2]

Question 4

- i) The formula of an oxide of a metal M is M_2O_3 . Write down the formula of its : [3]
 (a) Sulphate
 (b) Fluoride
 (c) Phosphate

- ii) Study the following graph of volume versus temperature given below and answer the following questions:



- (a) What is the volume of the gas at 300K?
 (b) At what temperature will the gas occupy 250 ml?
 (c) State the law represented by the above graph. [3]
- iii) Bromine occurs in nature in the form of isotopes $^{79}_{35}\text{Br}$ (49.7%) and $^{81}_{35}\text{Br}$ (50.3%), calculate the average atomic mass of a bromine atom. [2]
- iv) How would you distinguish between the following: [2]
 (a) Carbon dioxide and sulphur dioxide (using acidified potassium dichromate solution)
 (b) Chlorine and hydrogen chloride (using moist starch iodide paper)

Question 5

- i) Name the following: [3]
- (a) A gas which acts as an oxidizing as well as a reducing agent.
 - (b) A carbonate which does not decompose on heating.
 - (c) A compound which on heating decomposes to produce nitrogen as one of the gaseous product.
- ii) Elements X, Y and Z has atomic numbers 6, 17 and 12 respectively. Which one: [3]
- (a) Will gain one electron to achieve the octet electronic configuration?
 - (b) Has four electron in its valence shell?
 - (c) Has a tendency to achieve the electronic configuration of neon?
- iii) Hydrated calcium sulphate $\{CaSO_4 \cdot xH_2O\}$ contains 21 % water of crystallization. Calculate the value of x in hydrated calcium sulphate. [2]
- [Ca=40, S=32, O=16, H=1]
- iv) Draw the orbit diagram of the structure of the following atoms : [2]
- (a) Potassium
 - (b) Silicon

Question 6

- i) Correct the following statements by adding suitable word/s. [3]
- (a) Ammonia gas turns red litmus paper blue.
 - (b) A brown precipitate is obtained when sodium hydroxide solution is added to a solution of ferric sulphate solution.
 - (c) Oxygen turns pyrogallol solution brown.
- ii) With respect to the industrial preparation of hydrogen answer the following: [3]
- (a) Write the chemical equation for the production of water gas.
 - (b) Write the chemical equation for the catalyzed reaction.
 - (c) How is hydrogen separated from unused carbon monoxide?
- iii) Complete and balance the following ionic equations and classify them into oxidation or reduction reaction : [4]
- (a) $Mn^{5+} \rightarrow Mn^{3+}$
 - (b) $Cl^- \rightarrow Cl_2$.
 - (c) $Cu \rightarrow Cu^{2+}$
 - (d) $O_2 \rightarrow O^{2-}$

Question 7

- i) 100 ml of nitrogen collected at 27°C and 720 mm Hg is cooled to -73°C under a pressure of 760 mm Hg. What is the volume occupied by the gas? [3]

ii) Write balanced equations for the following conversions:

[3]

- (a) Copper carbonate to copper oxide
- (b) Lead to sodium plumbite
- (c) Silver chloride to silver

iii) With respect to the equations given below answer as instructed:

[4]

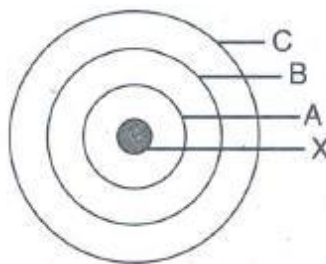
- (a) $\text{Fe}_2\text{O}_3 + 3\text{CO} \rightarrow 2\text{Fe} + 3\text{CO}_2$ (identify the oxidizing agent)
- (b) $\text{Cr}_2\text{O}_3 + 2\text{Al} \rightarrow \text{Al}_2\text{O}_3 + 2\text{Cr}$ (identify the substance undergoing reduction)
- (c) $\text{MnO}_2 + 4\text{HCl} \rightarrow \text{MnCl}_2 + 4\text{H}_2\text{O} + \text{Cl}_2$ (identify the reduced product)
- (d) $\text{H}_2\text{S} + \text{Cl}_2 \rightarrow 2\text{HCl} + \text{S}$ (identify the oxidized product)

Question 8

i) In the given diagram of atom Q, shell C contains one-fourth of the number of electrons present in shell B :

[5]

- (a) What is the atomic number of Q?
- (b) Write the electronic configuration of Q?
- (c) If the number of neutrons present in Q is 14, what is the mass number of Q?
- (d) State whether Q forms a cation or anion?
- (e) Write the formula of the compound formed between Q and oxygen.



ii) A small piece of potassium metal is put into a small trough containing water.

[5]

- (a) Name the gas formed in the reaction.
- (b) Give one chemical test for the identification of the gas.
- (c) Write an equation for the reaction taking place.
- (d) State the type of reaction taking place.
- (e) State **one** relevant observation for the reaction between potassium and water.